GROW computing Weeks 7 and 8

Two lessons of 40 minutes · Scratch maze completion

These lessons complete the eight-session core course. Pupils add one feature that changes gameplay, then test the whole game against clear expectations and improve it. Use a pupil's own saved project as the starting point each week.

Week	Shared goal	AQA focus
7	Add one challenge	Outcome 5
8	Test and improve	Outcome 6 and complete review

The lesson chassis

Stage	Minutes	Purpose
Arrival	0-3	Settle and locate the current file.
Starter	3-6	Retrieve and predict.
I Do	6-11	Watch a precise model.
We Do	11-18	Investigate and test together.
I Do 2	18-21	Model the next small part.
We Do 2 · Task	21-25	Rehearse the first independent action.
Independent Task	25-35	Create, test and explain an individual result.
Exit	35-40	Save, reopen and agree the next step.

Resources and access

Each week includes HTML, PowerPoint, a Main task booklet, Step by step booklet, Challenge booklet, teacher guide, PDFs and Scratch files. Choose and change routes from observed need. Quiet participation and spoken or pointed answers work in every route.

Evidence and time

Pupil creation is observed individually. Recovery files support access but cannot establish authorship of inherited content. The UAS lead records outcomes on AQA summary sheets. Week 8 reviews all six outcomes; a partly finished game receives a recorded next step, not an assumed completion.

This pack teaches a moving Hazard and contact rule. A collectible or scoring system is a possible later extension. Agree extra teaching time and complete instructions before offering it. Revisit the Week 6 readiness review where needed.

Current AQA unit 71638 Programming with Scratch unit 6 Level One

Week 7 Add one challenge

One 40-minute lesson

I can add one new feature that changes how the game is played.

Words: mechanic, hazard, glide, test

Stage and time	Teaching and pupil action
Arrival 0-3	Open your own W06_Win.sb3. Save a copy as W07_Challenge.sb3. Keep the working game safe.
Starter 3-6	A new background only looks different. A moving obstacle changes the route. Which one changes what the player must do?
I Do 6-11	Create a new sprite and name it Hazard. Give it two glide blocks inside forever. Put both glides inside forever. Keep the route clear of the start.
We Do 11-18	Player waits at x 0, y 0. Hazard starts at y -100. Predict what happens after 1 second, then test. Repeat with Player at x 30, y 0. Explain the difference.
I Do 2 18-21	The consequence belongs to Player. Player checks touching Hazard? Contact returns Player to the safe start.
We Do 2 · Task 21-25	Test contact, then a route that avoids Hazard. Restart and watch the complete outward and return journey. Choose a size and position that leave a route to Finish.
Independent Task 25-35	Build a moving Hazard and a contact rule on Player. Test the changed game twice. Explain what the player must now do.
Exit 35-40	Save as W07_Challenge.sb3. Load it and show the new feature. Explain the extra challenge.

Step by step: Build a moving Hazard and its contact rule, one step at a time.

Main task: Build both scripts. Test contact, avoidance and a fresh run.

Challenge: After the core task, compare 2-second and 4-second glides.

Voice: Revisit your Week 2 mission idea. Which feature is achievable today? Agree an adaptation and say why.

Evidence: AQA 71638 outcome 5: Observe the pupil creating the feature and explaining what the player must now do. Record a before-and-after test. Decoration alone does not meet the outcome.

Prepare: Open your own W06_Win.sb3 and save a copy. Have the named booklet, a teacher demo and the approved save location ready.

Week 8 Test and improve

One 40-minute lesson

I can test my game against clear expectations and explain one improvement.

Words: expected, actual, fault, retest

Stage and time	Teaching and pupil action
Arrival 0-3	Open your own W07_Challenge.sb3. Put Player at the safe start. Prepare your test record.
Starter 3-6	“It is broken” does not help. “The down arrow moves Player up” does help. A fault report names the test and the result.
I Do 6-11	Reset Player to the safe start. Predict one down-key press, then test. Record the position you actually see.
We Do 11-18	Share the named test setup before each check. Record the result, including passes. Choose one cause to inspect.
I Do 2 18-21	Change y by 10 to change y by -10 on the down-key script. Repeat the same down-key test. Then check the other controls and game rules.
We Do 2 · Task 21-25	Choose a peer or teacher tester. Agree a quiet way to respond. Ask which check they ran and what happened. If all checks pass, test a clearer instruction or harder case.
Independent Task 25-35	Run the five checks on your own game. Make one useful improvement. Retest the same steps.
Exit 35-40	Save as W08_Final.sb3. Show your test record and a repair, improvement or agreed next step. Review your six-outcome record.

Step by step: Use the five-check record. Say or point to each result.

Main task: Run all five checks. Record results and retest any change.

Challenge: After the core checks, investigate a corner or contact during a glide.

Voice: What did your tester's feedback change? Name the actual improvement, or explain why a suggestion was deferred.

Evidence: AQA 71638 outcome 6: Observe the pupil testing the game, recording expected and actual results, and retesting after a repair. Review all six outcomes individually before any award claim.

Prepare: Open your own W07_Challenge.sb3 and the bug project. Have the test record, the named booklet and the approved save location ready.